

Sample SOP — 5G Band 78 Indoor Deployment

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1. Scope

This section covers indoor small-cell deployment parameters for 5G NR operating in Band 78 (3.3–3.8 GHz), applicable to enterprise and public indoor venues managed by NCS field engineering teams.

2. Maximum Transmit Power — Indoor Deployment

For indoor small-cell units deployed under this SOP, the maximum permitted Effective Isotropic Radiated Power (EIRP) is 24 dBm per 20 MHz channel, in line with regulatory limits for indoor unlicensed-adjacent shared spectrum use. Field engineers must not exceed this threshold during commissioning, regardless of vendor default configuration values, which are sometimes set higher for outdoor macro deployments.

Note: Outdoor macro-cell EIRP limits differ substantially (up to 65 dBm depending on antenna configuration) and are covered in SOP NCS-5G-SOP-009. Do not apply outdoor power settings to indoor equipment.

3. Antenna Placement Guidelines

Indoor small cells should be mounted at a minimum height of 2.4 metres from finished floor level, with a minimum clearance of 0.5 metres from any metal structural element to avoid signal reflection interference.

4. Commissioning Checklist

Before marking a site as commissioned, field engineers must confirm: (a) EIRP measured at the antenna port does not exceed 24 dBm, (b) VSWR reading is below 1.5:1, (c) backhaul link latency is below 10ms to the nearest aggregation point, and (d) all commissioning readings are logged in the NOC ticketing system with site ID and timestamp.

5. Common Misconfiguration Risk

The most frequently observed field error is applying an outdoor macro-cell power profile to an indoor small-cell unit, resulting in EIRP readings of 40–45 dBm — well above the 24 dBm indoor limit. This SOP exists specifically to prevent that error during commissioning.

6. Revision History

v3.2 (15 March 2026): Clarified EIRP limit units and added common misconfiguration note. v3.1 (2 Nov 2025): Updated antenna clearance guideline. v3.0 (1 June 2025): Initial release for Band 78 indoor deployments.